

AI Edge REU Final Report

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Throughout the 2 month REU program, I have learned a lot about not only the application of AI into various fields such as wireless networks for better traffic routing, but also a lot about the processes that go behind the scenes, especially the effects of different parameters on training various different types of models. Still being in high school, a lot of the presented mathematical concepts were quite difficult to grasp, with difficulty ramping up by a considerable degree after the first two weeks. Regardless, the assignments were posted in such a sequence that I was still able to understand the inner workings of many different models and their applications. I feel that with a little bit more mathematical ability under my belt, a lot of the content of the program will fit into place, and then I will realise the true value of this experience. I learned a lot about what kind of tasks that AI is used for, those being tasks involving classification and drawing parallels between different items and situations. This definition is quite broad, but to give an example, AI is much more suited towards language generation and image tracking rather than, for example, ethics and judgement. Going into more detail about the effects of different parameters on model training, which is a topic that I had no prior knowledge in, one of the key takeaways that I got from this program was the importance of a balanced learning rate. Prior to this, with the limited knowledge that I had, I immediately assumed that training an AI model for longer periods of time would yield better results, and while this is generally true, I found out that with a poorly calibrated learning rate, the model could both overfit and underfit the loss curve and end up with poor output, irregardless of training time. I also learned a lot about my area of interest, which was large language models, through the Shakespeare text generation assignment. Especially interesting to me was the process of tokenization and embeddings, which showed me how

generative models break down connections (semantic, grammatical, etc) between words in order to form more coherent text generation. These were only a few examples, but I would say that, even for wholly inexperienced high schoolers, this experience is well worth the challenging concepts it presents.

The paper I decided to read was ‘What is ChatGPT Doing, and Why Does it Work?’, as it pertains to my field of interest, and has a lot of intersection with another field that I also have experience in, which is computational linguistics. So how does ChatGPT work? At the most basic level, ChatGPT works by predicting the next most likely word to come in a sequence of words, or token in a sequence of tokens. At each step of the text generation process, ChatGPT generates a probability distribution, and uses a word from the generated sample. In order to produce more human text, there is also a randomness factor called temperature, which varies the word choice selected from the generated sample. These probabilities come from a variety of different sources, but most common are letter frequencies derived from usage rates of letters in English orthography, n-grams of letter combinations, and on an even higher dimension than that, n-grams of entire words and phrases. This then begs the question, how are words selected for the incremental samples of text generation? The answer lies in tokenization, and embeddings & vectorization. In tokenization, the input is broken down into smaller units that are more suited towards processing, which helps the model analyze and break down the information presented to it. Embedding then converts the tokens into high-dimensional vectors, which can then be compared to other vectors in order to find the next most likely word. What is then considered as “next most likely” is a result of training off of text datasets. At its simplest, in ChatGPT’s training, a loss function must be defined in order to update the model’s output over time by

looking at “how correct” the output is, and adjusting weights and parameters in response, with the end goal of having a more accurate output in the next training loop. Through this constant feedback, the model is able to generate internal connections such as grammar, context relevant word choice, tone, and other general language patterns, about the input training data without having explicitly defined it to the model beforehand. This readies the model for fine tuning and reinforcement learning, which can include human feedback. This idea of information and semantics is then expanded upon through the realization of different meaning vectors in space, and by looking at how the model classifies certain words together, based on the characteristics of the use cases of the words. Summarized, ChatGPT works by training its neural network on training data, forms connections between words from the bottom up, quantifies the similarities through vectors, and outputs text based on continuously generated probability sets.

Looking at this paper, I would say that it thoroughly reflects what I have learned in the AI Edge program, and thanks to the experience that I got actually training some models, I was able to make sense of the terminology presented in the paper. It prompted me to wonder more about potential overlap between linguistics and generative AI, particularly where semantics and computational linguistics are concerned. It also made me realize how at a higher level, language and math are intimately connected, and also answered a few observations I have had regarding affinity between mathematical and linguistic capabilities in people.